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Techno-economic assessment of offshore wind and hybrid wind–wave farms with energy storage systems

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ABSTRACT

Ocean renewables (such as offshore wind and wave) are abundant and essential energy resources for supporting future emission-free targets. However, their energy intermittency and high cost have hindered commercialization and wide-scale implementations of these ocean energy technologies. This paper focuses on both issues and aims to increase the dispatchability of ocean energy farms by investigating the potential of a hybrid wind and wave energy platform with energy storage systems (ESS). This research introduces an innovative method to assess the necessary ESS capacity for offshore renewable energy farms, ensuring energy dispatchability to the local demand. In addition, the key criteria and the economic feasibility of deploying ESS in offshore energy farms are also discussed. A life-cycle cost model is presented that quantitatively compares different farm configurations, specifically a standalone offshore wind farm and a hybrid configuration incorporating both wind and wave generation systems. The results indicate that the combined wind and wave energy farm significantly reduces the ESS requirement and provides competitive lifecycle costs compared to the stand-alone wind energy farm, though the amount of these benefits vary on the local resource characteristics. The quantitative assessment presented in this study also provides realistic and practical justifications for combined energy farms that complement statistical conclusions derived from variability indexes in previous studies.

Abbreviations

AC Alternating current

AEMO Australia Energy Market Operator

CAES Compressed air energy storage

CAPEX Capital expenditure

DC Direct current

DCPEX Decommissioning expenditure

ESS Energy storage system

FES Flywheel energy storage

LCC Life-cycle cost

LCOE Levelised cost of energy

NaS Sodium–sulfur battery

O&M Operation and maintenance

OPEX Operational expenditure

PHS Pumped hydro storage

RES Renewable energy system

SOC State of charge

SS Support system

TRL Technology readiness level

VRFB Vanadium redox flow battery

WEC Wave energy converter

WT Wind turbine

1. Introduction

Offshore wind and wave renewable energy sources have great potential; hence they are likely to play a critical role in forming the energy supply landscape of the future in the countries with sea/ocean borders and considerable offshore energy availability. Compared to onshore wind farms (which are almost saturated), offshore wind farms can provide significant benefits in harvesting much more stable and stronger wind resources pushing the wind turbine (WTs) sizes to their limits [1]. In addition, offshore wind farms also minimize visual pollution and eliminate the noise impacts on the local community.

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Globally, offshore wind energy has grown significantly over the past decade as the installed capacity has increased by 21% each year since 2013. In addition, it is predicted that about 370 GW offshore wind will be installed by the end of 2030 [2]. This shows an increase of 85% compared to a prediction of 2021 (200 GW by 2030 [3]). Furthermore, it is predicted that the share of offshore wind energy in global new wind capacity will rise from 23% in 2021 to 30% by 2031.

Wave energy is another ocean renewable resource having greater energy generation potential and higher predictability over wind energy [4,5]. However, unlike WTs (which have technological maturity and displayed significant growth within the last two decades), wave energy converters (WECs) are not commercially viable yet though a range of devices has been proposed in the last century [4]. Therefore, there is an opportunity to support the development of WEC systems that can offer competitive solutions to complement the mature WT technologies.

Although WEC technologies are still immature in general, it is observed that the topology of the WEC systems has gradually converged over the 20 years into three main types [6]: Oscillating Water Column, Oscillating Bodies and Overtopping devices, which have already demonstrated significant potential in commercialization via sea trials across the globe.

It is foreseen that two primary challenges are impeding the development of ocean renewables: high capital expenditures (CAPEX) and power intermittency. The former is due to lack of convergence of WEC designs to a commercial product and the technical challenges of deploying WTs in deep sea applications (which require floating structures and complex controllers). The latter issue, power intermittency, however, is connected to the nature of the ocean renewable energy resources — larger power potential presents large power variations in offshore wind and wave resources. This leads to significant challenges in the effective transmission of power to the electricity market and the mitigation of power intermittency on the grid operation and security.

As the power system transition progresses globally, energy storage systems (ESS) play a crucial role in mitigating the energy variability that exists in intermittent renewable sources, particularly to enhance the future electricity supply considering the retirement of fully dispatchable fossil fuel generations. In [7], an overview is presented on sizing methodologies applied for a variety of renewable energy systems (RES) with both generating and storing subsystems. In the field of wind farms, studies have focused on sizing only the ESS subsystem based on specific applications in connection to the renewable energy generation systems [8,9]. In addition, the exploitation of offshore renewables is anticipated on an enormous scale to achieve the future net-zero target. This endeavour requires an appropriate assessment of energy storage technologies that can be collocated with ocean energy farms. However, the technical and economic feasibility of ESS for offshore energy farms has not been systematically studied in terms of their feasibility and types, and the requirement of suitable ESS sizing to form a more dispatchable renewable energy supply has not been investigated in the literature.

The integration of offshore wind and wave energy is an emerging concept which could be an alternative solution to address these challenges through their unique benefits such as cost reduction due to the shared facilities [10–12] and energy variability reduction due to the complementary behaviour between wind and wave [13–15]. Other benefits of combined wind and wave include the improvement of platform stability [16] and enhanced energy production [13,17,18].

However, previous studies on combined wind and wave systems mainly focus on the hydrodynamic design [19–21], resources and location assessment [18,22,23] and the economic analysis/comparison [10, 24,25]. In terms of energy variability, most studies of combined energy farms focused on the modelling and analysis of various statistical variability indexes to prove that the hybrid system has unique advantages in energy variability reduction. However, such statistical assessments,

though these analyses are quantified, struggle to provide a straightforward and practical justification as it is hard to link their results with the technical and economical sizing of ocean energy farm configurations. In addition, energy variability is not fully equivalent to the energy dispatchability to the local energy market, where the latter involves multiple power systems factors [26,27], such as the generation and demand balance, economic dispatch optimization, curtailment and power system constraints. However, the dispatchability of combined offshore energy farms and the unique advantages to the local demand are not fully investigated and compared with standalone offshore wind farms in the previous studies.

To address these gaps in offshore energy farm configurations, this study introduces an innovative approach for estimating the requisite Energy Storage System (ESS) capacity in diverse offshore configurations, emphasizing their energy dispatchability to meet localized demand. In addition, the key criteria for deploying ESS configurations in offshore farms are discussed. Furthermore, this study develops a comprehensive and high-fidelity life-cycle cost model to thoroughly examine the techno-economic viability of different offshore farm setups when incorporating ESS options. This research makes two significant contributions. Firstly, it conducts a systematic techno-economic assessment that comprehensively explores the feasibility of various ESS alternatives for offshore energy farms. Secondly, it quantifies the benefits of dispatchability in combined energy farms from both technical and economic perspectives, providing realistic and practical justifications that complement statistical conclusions derived from variability indexes. Moreover, the sensitivity of these combined energy farm benefits to location-specific factors is investigated, comparing two sea sites with distinct resource characteristics in the Southern and Northern hemispheres, offering valuable insights into the geographical variations' impact on ESS-integrated combined energy farm feasibility and advantages.

This paper is structured as follows: Section 2 analyses the wind and wave data statistics and introduces the methodology used. The applied economic cost model for offshore energy farms with different system configurations is described in Section 3. Section 4 investigates the main results. The techno-economic feasibility of EES options under distinct location characteristics are discussed in Section 5, followed by the conclusion.

2. Methodology

2.1. Offshore locations and wind and wave resources

For determining suitable sites for combined offshore wind and wave energy farms, several approaches have been proposed in previous studies, such as the wave system identification method and site selection matrix in [18] and the co-location feasibility index in [23]. In this paper, two offshore sites, one near Sydney (152.31°E, 34.00°S) in Australia and the other in the North Sea (8.03°E, 56.65°N) in Europe, are selected as the case studies. Note that these locations have been reported to be the optimal locations for developing combined wind and wave farms in their studied region [18,28].

The case study in Sydney uses the wind-wave hindcast data (obtained by the WaveWatch III model) running on a series of nested grids with 7 km resolution in the Australian region [29]. The data used in the North Sea site is obtained from ERA5 [30], which is hourly data covering long historical wind and wave data from 1940 to the present day. The main parameters considered in this paper include the wind speed at 10 m height (U_{10m}), significant wave height (H_s) and wave peak period (T_p). Since wind generation power is proportional to the cube of the wind speed at hub height, the power in the wind is significantly affected by wind speed variation (the impact of the height and the roughness of the blowing surface). Thus, the wind speed at hub height (U_{100m}) is calculated by:

$$U_{100} = U_{10} \cdot \left(\frac{z_{100}}{z_{10}} \right)^{\alpha}, \quad (1)$$

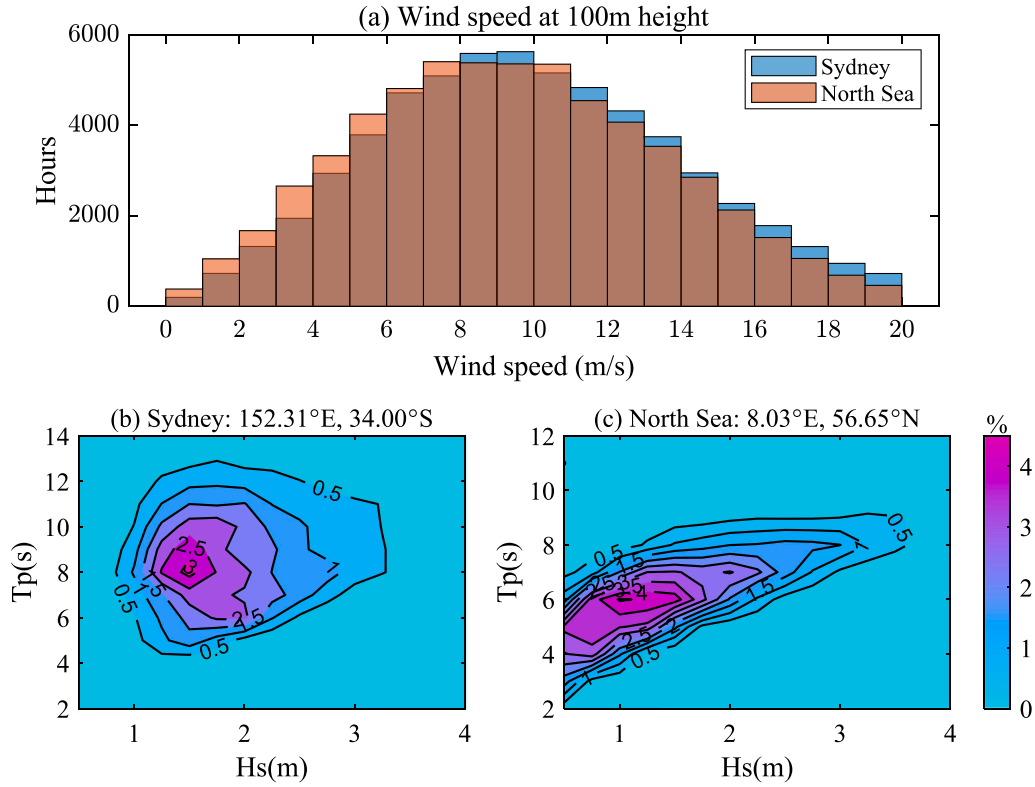


Fig. 1. Wind and wave resources in Sydney and North locations from 2014 to 2020 [29,30]: (a) wind histogram; (b) wave scatter plot in Sydney; (c) wave scatter plot in the North Sea.

where α is the friction coefficient with a value of 0.1 under open water terrain [31].

Fig. 1a shows the wind speed histogram of two sites at 100 m height, indicating that both locations have good wind resources with the majority of wind speeds between 7 and 12 m/s (IEC Class II or above [32]) and the Sydney site has a slightly higher wind resource compared to the North Sea site. In terms of wave parameters, Figs. 1b&c show the wave statistics scatter map in the Sydney and North Sea sites. Significant discrepancies can be identified for the sea states in Sydney that are more concentrated around $H_s = 1.5$ m and $T_p = 8$ s, while for the North Sea, the most frequently occurring sea state is around $H_s = 1$ m and $T_p = 6$ s with a wider distribution of H_s . According to the wave system identification in [18], the shorter wave peak period at the North Sea location suggests that the waves are mainly generated by the wind with a minor fraction of swell waves. This may lead to a higher correlation and shorter lag time between wind and wave resources.

2.2. Method of power generation

The wind power generation is calculated by the power curve of a commercial WT (Gamesa G128-5MW [33]) that is given as a function of the wind speed at hub height as shown in Fig. 2 (top). This turbine is suitable for deployment both in onshore and offshore farms [34]. The wave power generation is estimated by the power matrix (Fig. 2 bottom) of the WaveStar C6 600 kW WEC at given significant wave heights (H_s) and peak wave periods (T_p). WaveStar C6 WEC was chosen as it has been commercially and experimentally tested and recognized as a relatively mature technology (TRL 7) as reported in [35–37]. To avoid intensive simulations, the power curve and power matrix of the WT and WEC are assumed to be consistent for different deployment sites. In addition, this study assumes that the WT with an ideal yaw system and WEC with axisymmetric buoys can capture wind and wave resources from all directions, and the water depth and support structure have a

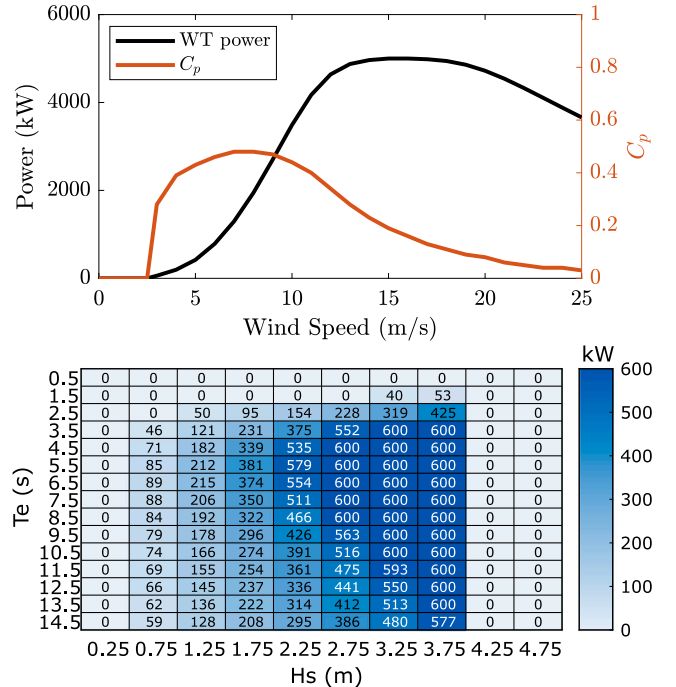


Fig. 2. Power curve of Gamesa G128 5 MW wind turbine (top) and power matrix of WaveStar C6 600 kW WEC (bottom).

limited impact on their power generation, although it is acknowledged that the direction of incident wind and wave and supporting structure may affect the generation performance of both technologies.

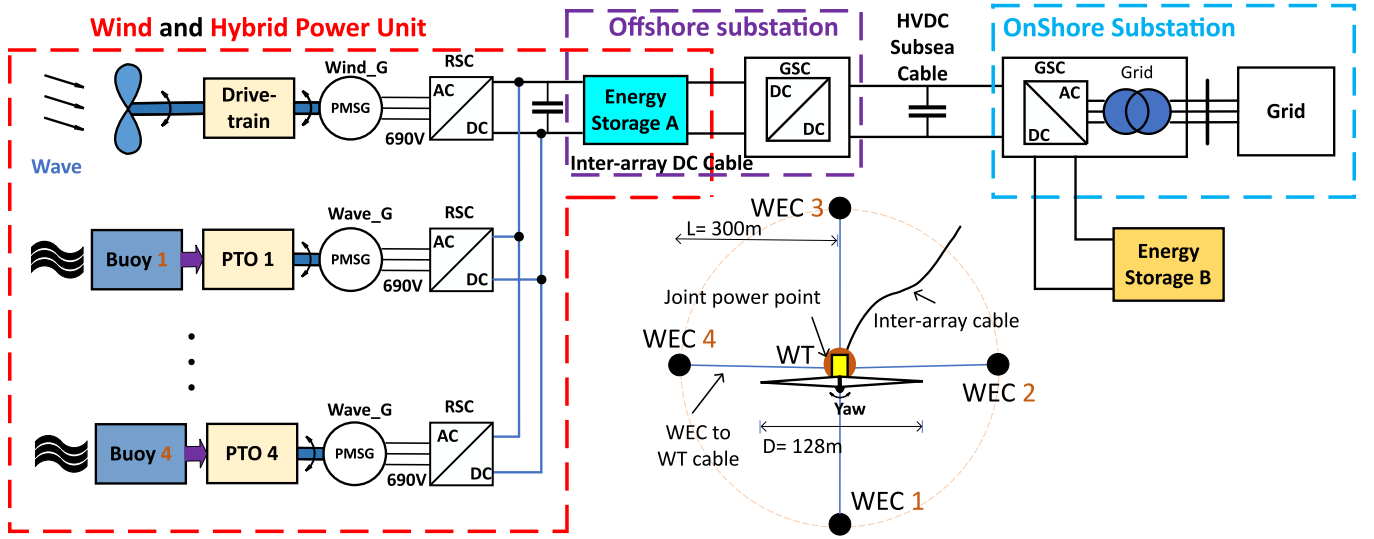


Fig. 3. A diagram of the hybrid offshore wind and wave conversion system.

2.3. Hybrid system configuration

In this work, a DC-linked hybrid offshore wind and wave conversion system is used to perform the techno-economic analysis, as shown in Fig. 3. The configuration of offshore hybrid offshore devices comprises a wind turbine coupled with 4 WECs surrounding the wind turbine in an axisymmetric layout, as illustrated in the bottom-right of the diagram. The four WEC devices are evenly distributed around a WT with a distance of 300 m, allowing 424 m distance (much more than 4 radii) between WECs [38] to minimize the interactions between WEC devices within one combined unit. Note that such a combined unit can be duplicated and distributed at a farm scale. However, the interactions between combined units are assumed to be negligible in this work, considering sufficient distances in the farm design [15,25] as the specific geographical farm design is not investigated in this research. The direction of WEC arrays is determined by the assessment of the maximum intermittency reduction based on the incident wave direction of each sea state [39]. The block diagram around the schematic shows the conversion systems of the wind and WECs (in the red-dashed block), the collocated energy storage systems (such as the offshore option A and onshore option B), the DC-DC converter with high-voltage DC transmission line on offshore substation (in the purple-dashed block) and the onshore substation (in the blue-dashed block).

Note that the DC system in a hybrid power unit presents a number of benefits. These include facilitating the integration with various DC converters running at various system frequencies from multiple generator units. Note that a direct drive (no gearbox) brushless permanent generator is considered in the offshore wind turbine system since it is the preferred generator technology as it offers a highly reliable system that is desirable in a harsh operating environment. In addition, DC conversion is preferable to be able to integrate multiple WECs that will operate under different conditions. High voltage DC sub-sea cables are also considered since they are suitable for long-distance transmission (typically above 50–80 km [40], such as in the Sydney location).

It is also noted that there are two potential energy storage options: Energy Storage A and Energy Storage B (in Fig. 3). The option of Energy Storage A can be deployed distributively on each hybrid/WT-alone platform, or it can be a large unit centralized on an offshore substation. On the other hand, the Energy Storage B option performs as a normal onshore energy storage station. The technical and economic comparisons of these two storage options are investigated in Section 5.

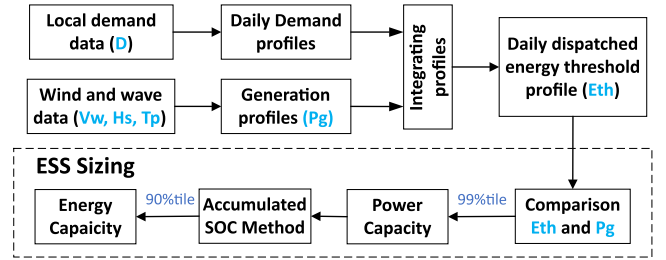


Fig. 4. Method of dispatching energy with statistical sized ESS.

2.4. Managing power dispatchability using energy storage systems

It is known that energy storage systems have been widely applied to reduce the variability and intermittency of non-dispatchable renewable generation. In this paper, the methodology of managing the variability and intermittency of offshore renewables by determining the suitable size of energy storage is proposed, as illustrated in Fig. 4. This effective algorithm firstly generates the daily demand profile and generation profile by utilizing the real demand data as well as the wind and wave data. These profiles are integrated to obtain the energy dispatch profiles and then compared with generation time series data to calculate the energy difference (deficit and surplus). Finally, the energy deficit and surplus are used to size the ESS capacity by the accumulated state of charge (SOC) method.

The demand information is obtained from the Australian Energy Market Operator (AEMO) [41] and from the European Network of Transmission System Operators for Electricity [42]. The normalized daily demand profile (DP^i , a vector of 1×24) at day i can be defined by:

$$DP^i = \frac{(D^i - \bar{D}^i)}{\bar{D}^i} \quad (2)$$

where D^i is the hourly demand data at day i and $\bar{D}^i$ is the mean value of demand data at day i . The dispatch energy threshold E_{th}^i at day i is determined based on the mean value of the generation profile ($\bar{P}_g^i$) at day i and the normalized demand profile at the same day, according to

$$E_{th}^i = \bar{P}_g^i \times DP^i. \quad (3)$$

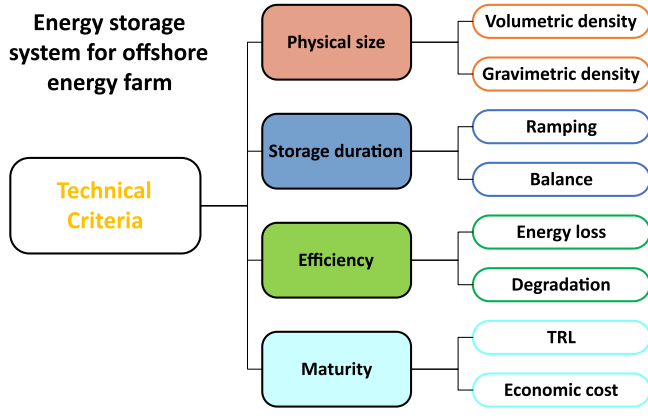


Fig. 5. Technical criteria of deploying ESS in offshore energy farms.

The power capacity can be estimated by the energy deficit and surplus (ΔP_g), which is the energy difference between the power generation time series (P_g) and the energy dispatch threshold time series (E_{th}) in a year, given by:

$$\Delta P_g = P_g - E_{th} \quad (4)$$

where $E_{th} = [E_{th}^1, \dots, E_{th}^{365}]^T$ and $P_g = [P_g^1, \dots, P_g^{365}]^T$. For best practice, seven years generation data (from 2014 to 2020) in the Sydney location is used to estimate power capacity. The charging/discharging rated powers (P_c and P_d) of the ESS are assumed to be equal and estimated to be the average of the 99th percentile of ΔP_g of each year to cover most of power charging and discharging event.

The corresponding ESS energy capacity is determined by the distribution of the accumulated SOC series [13], which is defined by:

$$S_t = \begin{cases} \min(\Delta P_g, P_c) & \text{if } \Delta P_t > 0 \\ \max(\Delta P_g, P_d) & \text{otherwise,} \end{cases} \quad (5)$$

and

$$A_t = \begin{cases} S_0 & \text{if } t = 0 \\ A_{t-1} + S_t & \text{otherwise,} \end{cases} \quad (6)$$

where S_t and A_t are charging or discharging power series and accumulated SOC series, respectively. Note that, as the initial values of S_0 present a significant impact on the accumulated SOC data, the first step is to determine its initial values, which are given by $S_0^y = -\text{mean}(A_t^y)$ to balance the offset of the SOC data series at each year y . Then, the ESS energy capacity is estimated by taking the average of the 90th percentile of A_t^y at each year between 2014 to 2020.

2.5. Evaluation of technical criteria of ESS

After assessing the necessary ESS size based on demand and production, several critical technical features are considered for determining the ESS technologies for offshore energy farms. Fig. 5 shows four key technical criteria proposed in this paper to evaluate ESS types, including the physical size, storage duration, efficiency and technical maturity. The volumetric and gravimetric power and energy density are two technical parameters to determine the physical size of ESS for an offshore energy farm. The level of storage duration is the amount of time an ESS can store and discharge energy before needing recharging, and it is determined by the application's need for ramping and balance of energy demand. Energy efficiency and technology maturity are two important factors as they are highly related to energy loss, degradation (lifetime) and cost.

Note that all these factors are used to assess the feasibility of the different ESS types and choose an ESS for the specific context or farm

configurations. The optimal size and economic features are specifically investigated in this paper, while the other criteria are evaluated by assessing previous literature reviews focusing on the technical characteristics of different ESS types. However, the weight given to each feature for a complete assessment of the ESS types depends on the stakeholders' requirements, the specific characteristics of a site, and the power network features.

3. Economic model

From a financial point of view, the different RESs can be compared through an approximation of their life cycle costs (LCC) and energy generation over their lifetime. The most commonly applied metric for a techno-economic assessment of energy generation plants is the Levelized Cost of Energy/Electricity (LCOE) which displays the net present value of the studied system by applying a discounting method on the costs and outputs of the RES. Different calculation methods of LCOE have been presented and compared in [43]. In this paper, the LCOE is defined by [25]:

$$LCOE = \frac{\sum_{t=0}^n C_t / (1 + r_{\text{discount}})^t}{\sum_{t=0}^n E_{\text{tot}} / (1 + r_{\text{discount}})^t} \quad (7)$$

where C_t and E_{tot} are the annual total cost and annual total energy production, respectively, and the variables n and r_{discount} are the lifetime of the RES (which is taken as 25 years in this paper) and the discount rate which reflects a complex quantitative analysis covering financial, social and environmental factors; it is taken as the average value of r_{discount} used in case studies in [44] and set to 7.6%. The annual energy production (E_{tot}) is estimated by averaging the 7 year generation data (calculated as in Section 2.2), given by:

$$E_{\text{tot}} = \frac{\alpha \eta}{7} \left[\sum_{t=2014}^{2020} E_t \right] \quad (8)$$

where E_t is the energy generation at year t ; α and η are the energy production availability (percentage of time for normal generation of energy farm), depending on the time window for maintenance operations [45], and the energy conversion system efficiency, depending on the total energy loss of the RES.

The annual total costs C_t are divided into capital expenditures (CAPEX), annual operational expenditures (OPEX) and decommissioning expenditures (DCPEX), covering the total life cycle costs of each subsystem of the studied RES, and they are given by:

$$C_t = \text{CAPEX} + \text{OPEX} + \text{DCPEX} \quad (9)$$

Note that CAPEX represents the initial investment costs before the RES is fully operational. The investment costs include the costs of: the development and consenting of the farm ($C_{\text{D\&C}}$), the building and purchasing costs (C_{Build}), the installation and commissioning (C_{Install}) and the power connection and balance of plant ($C_{\text{Connection}}$) for the offshore energy farm, as given by:

$$\text{CAPEX} = C_{\text{D\&C}} + C_{\text{Build}} + C_{\text{Install}} + C_{\text{Connection}} \quad (10)$$

The OPEX includes the operation and maintenance ($C_{\text{O\&M}}$), insurance ($C_{\text{Insurance}}$) and administrative (C_{Admin}) costs, given by:

$$\text{OPEX} = C_{\text{O\&M}} + C_{\text{Insurance}} + C_{\text{Admin}} \quad (11)$$

To categorize the costs of the different subsystems occurring at different project periods, the life cycle costs (LCC) of the entire project are structured as shown in Fig. 6. Overall, the LCC can be grouped into three subsystem costs: the generation system (GENC), energy storage system (ESSC) and supporting system costs (SSC). The GENC covers the costs of all the generation systems in the offshore energy farm, such as the WT, offshore foundations and/or WEC systems; the ESSC comprises all the costs related to the ESS; the SSC includes the costs of power connection, such as onshore substation, offshore substation

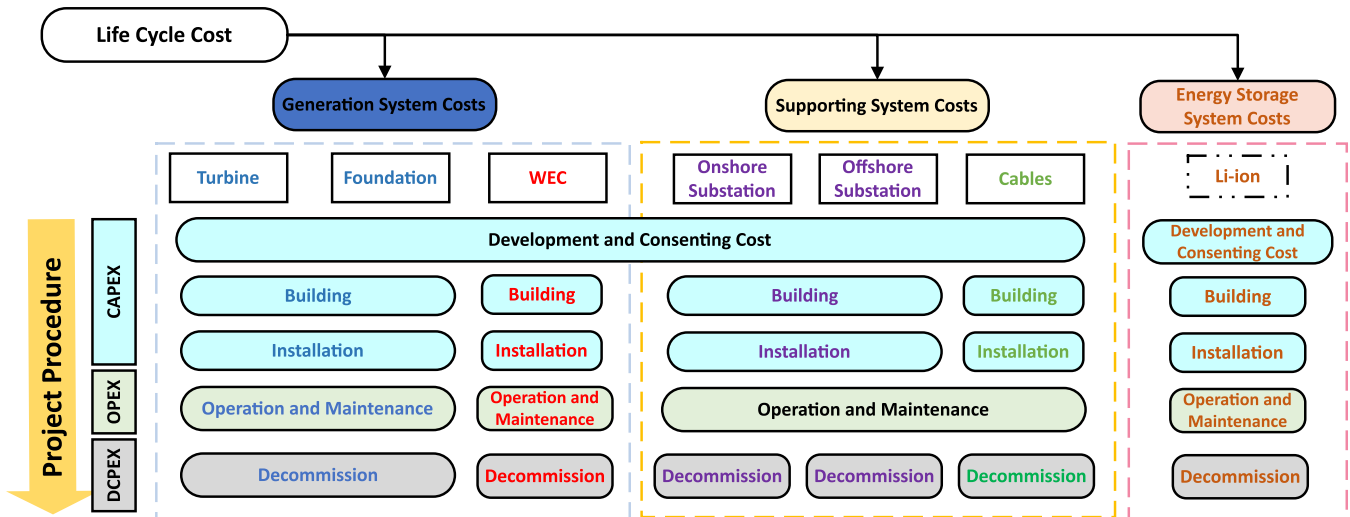


Fig. 6. A diagram of the lifecycle cost model is categorized: (1) by different project periods: CAPEX in cyan, OPEX in green, and DCPEX in grey; (2) by different subsystems of the RES: the generation systems (such as wind turbine in blue and WEC in red), the supporting system (such as the offshore substation in purple and the inter-array and connection cables in green) and the energy storage system in orange.

and cabling cost. In terms of the cost at different project development stages, the cost modules of CAPEX, OPEX and DCPEX are highlighted in cyan, green and grey respectively.

It is necessary to define key assumptions that were made in order to be able to financially assess and compare the different cases. As it is reported in [24,46,47], the costs of RES systems are project- and location-dependent. Especially for offshore wind farms, location dependent parameters such as water depth, distance to shore and infrastructure and resource availability influence the financial metrics considerably [48]. It is assumed that a company that has experience and knowledge in offshore projects will develop and install the hybrid system with the same approach and procedure in both locations [49]. The detailed analysis of three subsystems is described in the following subsections. All costs obtained from literature are converted to € and summarized in Table 1.

3.1. Generating systems costs

The cost characteristics for the WT generation system are taken from a technical report based on a 1 GW wind farm with 100 units of 10 MW in a location 60 km offshore in 30 m depth, which was planned to be operational in 2022 [47,50]. This technical report includes a more detailed breakdown of all CAPEX and OPEX costs; all costs are normalized with installed capacity (per MW), which is applied to the case study of this research.

For approximating the costs of the WEC units, an estimation of a generic, utility-scale floating wave energy farm with a 160 MW capacity is selected; its characteristics are summarized in [51]. This paper gathers more recent estimations of costs for the wave energy sector while keeping in mind influence factors such as learning curves and economy of scale of future projects [51].

When considering a hybrid system that shares infrastructure and is co-located, cost advantages can be justified. Especially, sharing significant investment costs such as development and consenting and connection costs (described in more detail in Section 3.2) leads to a decrease of 12% in total capital costs of the generation system [25]. Also $C_{O\&M}$ of the total generation system can be decreased by 12% as personnel and service vessels can be shared [25,45]. The hybrid system also increases the availability (α) and total energy generation of the WT units surrounded by WECs as it increases the accessibility to the WT for O&M tasks; an offshore wind farm shows $\alpha = 80\%$, a combined farm

reaches α of 90% for the wind turbines, while the α of WEC generation system is 95% and stays equal in both RES cases [45]. In our analysis, η is assumed to be 0.9 for both WT and WEC generation systems.

3.2. Supporting system costs

These RES-wide costs include the $C_{D\&C}$, offshore and onshore substation and the inter-array and export transmission cable costs (as depicted in Fig. 6). Such costs depend on the installed capacity of the generation system, the inter-array cable connection length (km_{ia} in Table 1) and the distance to shore, respectively. In this paper, all generating subsystems are assumed to be mature and at utility scale. Therefore, for the supporting system, utility scale costing parameters can be applied which are taken from the technical report of a 1 GW offshore wind farm [47,50]. The values for the supporting system CAPEX and OPEX are summarized in Table 1.

3.3. Energy storage system costs

Recent publications have focused on extensively comparing and reviewing possible ESS technologies, both technically and economically [52,53]. Parameters characterizing the technical aspects (e.g. power and energy capacity, density and lifetime) and the financial assessment (e.g. capital cost and LCOE) are represented using broad ranges of values as they are highly dependent on the maturity level of the ESS technologies selected for specific projects and their required capacity for the application [53]. A number of novel offshore – co-located or hybrid – storage systems have been introduced [54] but due to their low maturity level no extensive economic analysis is available, yet. Furthermore, the ESS does not influence the other RES costs (costs of generating and supporting system); both offshore and onshore storage solutions are considered as the generation farm is of utility scale. Although it is acknowledged that offshore ESS would increase the CAPEX and OPEX due to installation and operational costs compared to the similar onshore counterparts, this aspect has not been included in the financial assessment as limited data is currently available due to the absence of such an application. For ESS systems that have a lower lifetime than the defined project lifetime, the replacement costs are solely based on the CAPEX calculated with the energy capacity. Note that, unlike the energy storage unit, the power conversion unit in ESS usually does not have to be replaced. Li-Ion is chosen for the initial

Table 1
Financial input parameters of subsystems for the offshore RES.

System	Subsystem	Parameter	Value	Unit	Ref
GEN	WT	CAPEX	1789	k€/MW	[47,50]
		OPEX	63.8	k€/MW/year	
		DCPEX	139.2	k€/MW	
	WEC	CAPEX	3100.0	k€/MW	[51]
		OPEX	124.0	k€/MW/year	
		DCPEX	–	N/A	
ESS (Li-Ion)	Power Unit	CAPEX	372.5	k€/MW	[52–58]
	Storage Unit	CAPEX	357.1	k€/MWh	
		OPEX	9.6	k€/MW/year	
		DCPEX	–	N/A	
SS	Substation		398.5	€/MW	[47,50]
	Export Cables	CAPEX	6.8	k€/MW/km	
	Array Cables		203.0	k€/km _{ia}	
		OPEX	24.4	k€/MW/year	
	Substation		75.4	k€/MW	
	Cables	DCPEX	2.7	k€/MW/km	

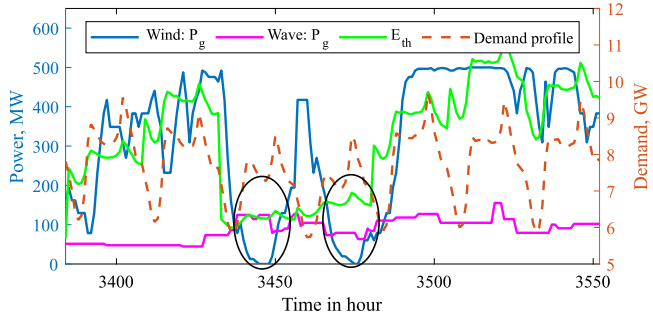


Fig. 7. Portion of time series (one week) of generation, dispatch threshold and demand profiles in the Site A.

case study to compare the systems in the two selected locations, as Li-ion ESS is widely applied and has reached a high level of technological maturity. It is noted that the applicability and economic feasibility of other ESS options are discussed in Section 5.

4. Results

4.1. Requirement of energy storage system capacity

According to the sizing method illustrated in Section 2.4, the ESS capacity required to meet the defined application is calculated in the sea site A (Sydney location) as an example. More specifically, Fig. 7 shows the portion of the time series of the power generation from wind (blue line) and wave (purple line), the energy dispatch threshold (green line) and the demand information (orange dashed line). It can be seen that the energy dispatch threshold profile follows the daily demand profile indicating that the commitment of the offshore energy farm to supply energy to the grid corresponds to the local demand. The difference between generation and the dispatch threshold is the requirement for storage, which is used to size the power and energy capacity of ESS. The black circles in Fig. 7 highlight the advantages of a combined wind and wave farm during no/low wind periods, as the dispatch threshold in those periods can be partially or even fully met by the wave energy generation instead of significantly relying on the storage capacity.

Regarding the power capacity of the ESS, Fig. 8 shows the hourly charging and discharging statistics. By taking the 99 percentile values into account, the power ratings of ESS for a stand-alone wind farm

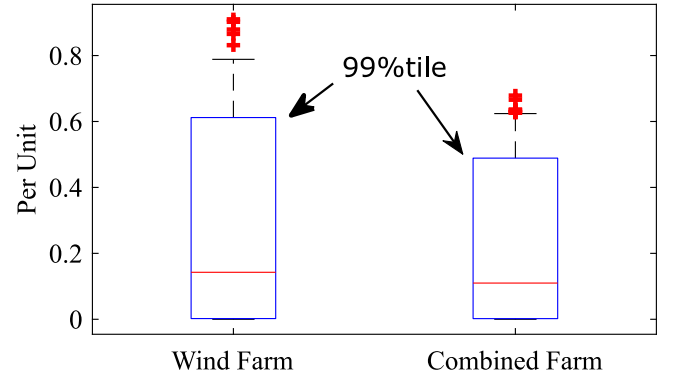


Fig. 8. Statistical power capacity for ESS in Site A.

and a combined energy farm in Site A are 0.6 per unit (p.u.) and 0.48 p.u. of the total farm installed capacity (corresponding to 300 MW and 240 MW, respectively). In addition, the ESS power capacity of the wind farm and combined farm for Site B (North Sea location) are estimated at 0.64 p.u. and 0.54 p.u. (320 MW and 270 MW, respectively).

Based on the accumulated SOC method (see Eq.), the ESS energy capacity can be estimated by averaging the yearly capacity A_t^y over 2014 to 2020. Overall, the energy capacity of ESS for a standalone wind farm and a combined energy farm in Site A are estimated at 1320 MWh and 864 MWh (equivalent to 3.6 h and 4.4 h of storage duration). These values in Site B are 1280 MWh and 1053 MWh (4 h and 3.9 h of storage duration, respectively). These results provide a quantitative assessment of the ESS requirements for two different offshore energy farms that highlight the unique advantages of combining wind and wave energy generation compared to a standalone energy farm in terms of the reduction of ESS requirement.

4.2. Site characteristics

Table 2 summarizes the geographic information such as distance to the shoreline (D_s) and water depth (D_e), the correspondence (where C_r and τ are the cross-correlation coefficient and lag time between wind and wave power) and power densities (P_{wind} and P_{wave}) of the wind and wave resources in two different locations. Note that comparing the distance to the shoreline and water depth of the locations is necessary to quantify the technical and financial parameters characterizing the subsystem of the substation and the supporting structure of the offshore wind turbine generation system.

It can be seen that site A (in the Sydney location) is a far-shore location with a 90 km transmission distance and deeper water of about 100 m, while site B (in the North Sea location) is a near-shore location having an 8 km distance to the shore and shallow water characteristics (with 35 m water depth).

The adjacent distance to the shore is the primary driver that influences the transmission cable length, hence defining the voltage level. This leads to higher supporting system costs. In general, AC transmission is more economically beneficial for distances below a specific threshold (such as 56 km in [40], 72 km in [59]). This is because the cost of a DC substation is higher than that of an AC substation, while DC cables are cheaper than AC cables. However, in this paper, the DC transmission cables are selected for both locations as the DC-linked hybrid system is used in this paper (as shown in Fig. 3), which eliminates the expensive DC/AC converter stations.

The water depth plays an important role in selecting offshore wind foundation types. Floating WT platforms can be deployed in sea sites with a water depth of more than 100 m, while bottom-fixed WT foundations are more economically feasible to install in a shallow water area with depths of less than 50 m [49]. Therefore, in this paper, the

Table 2
Resource and economic assessment of three system configurations in two locations.

Site	D_s/D_e km/m	C_r/τ -/h	P_{wind}, P_{wave} kW/m ² , kW/m	Configurations	ESS MW/MWh	Gen. GWh/yr	LCOE €/MWh
Site A	90/100	0.60/5	1.02, 21.6	WT only	–	26.0	93.6
				WT+ESS	300/1320	26.0	132.0
				WW+ESS	240/864	38.6	110.4
Site B	8/35	0.81/1	0.87, 13.3	WT only	–	24.4	84.7
				WT+ESS	320/1280	24.4	125.1
				WW+ESS	270/1053	33.6	118.0

floating platform and the bottom-fixed monopile foundation are used in Site A and Site B, respectively. The estimated costs of the two platform types are obtained from [60]. In the same reference, the monopile and the floating structure present a relatively comparable CAPEX value since the former has a higher cost of manufacturing, while the latter has a higher mooring cost. Note that in a more detailed cost structure for both foundation types the costs significantly depend on the soil characteristics of the selected locations; however, this is out of the scope of this paper. It is also noted that the key comparisons of different farm configurations are based on the same location and hence such influence would not impact the conclusions drawn by the presented results.

As illustrated in Section 4.1, for offshore energy farms with the same installed capacity (500 MW), the combined energy farm has lower requirements on both power and energy capacity of the ESS compared to the stand-alone wind energy farm (shown in Table 2). However, the benefits vary in the two distinct locations. The combined energy farm in Site A reduces the power capacity of ESS by 20% (from 300 MW to 240 MW) and the energy capacity by 35% (from 1320 MWh to 864 MWh), while in Site B power capacity of ESS is reduced by 15.6% (from 320 MW to 270 MW) and energy capacity is reduced by 17.7% (from 1053 MWh to 1280 MWh). Note that Site A has better performance in reducing both the ESS power capacity (20% versus 15.6%) and energy capacity (35% versus 17.7%), which is a result of a lower correspondence between wind and wave power. These results are consistent with the findings observed from the study in [18] that the lower correlation (C_r) and longer lag time (τ) are two preferable factors indicating better power smoothing (lower energy variability) as wind and wave resources complement each other. Therefore, the smaller energy deficit and energy curtailment from energy farms give rise to a reduced necessary size of ESS and hence lead to a preferable financial cost.

4.3. Economic analysis

The LCOEs of three energy farm configurations in the two locations are compared in Table 2. Although abundant wind and wave energy can be captured in Site A, the LCOEs of both the standalone WT farm and the hybrid WT with ESS farm (93.6 €/MWh and 132.0 €/MWh) are higher than those of Site B with values of 84.7 €/MWh and 125.1 €/MWh respectively, mainly due to the costly far-shore transmission. In addition, as a result of the smaller storage capacity and higher availability of resources in Site A, the LCOE of a combined energy farm with a Li-ion battery (as an ESS) has a considerably lower value of 110.4 €/MWh. The LCOE values of the wind-only farm with different turbine sizes, locations and future costing trends are studied in [46,50].

As illustrated in Fig. 9, the total investment costs of the generation system in the combined energy farm at the Sydney location show a slight increase of 8% compared to the wind-only farm due to the higher CAPEX of WECs. Note that the WT CAPEX in the combined energy farm significantly decreases due to fewer generation units being required to reach the same capacity (500 MW) of the energy farm and the shared infrastructure and installation costs.

In terms of ESS CAPEX, the combined energy farm significantly reduces the ESS CAPEX by 32% compared to the stand-alone wind

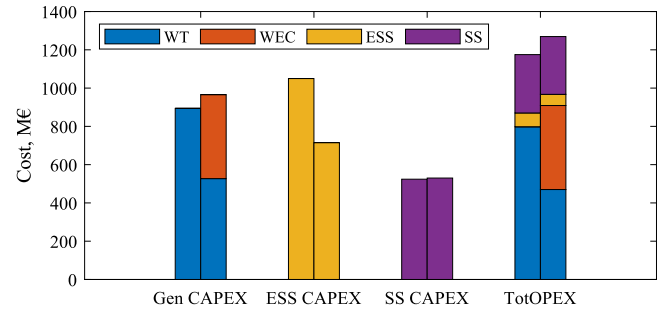


Fig. 9. Breakdown of undiscounted costs of 500 MW offshore energy farms in the Sydney location, under wind only (left groups) and WW system (right groups) configurations.

energy farm, highlighting the potential of combined wind and wave energies and their feasibility in future utility-scale projects. On the contrary, the SS CAPEX is slightly higher in the combined farm due to the additional cost of inter-array cabling between WECs and WT (as shown in Fig. 3). Note that the majority of overarching costs stay comparable due to coordinated efforts such as substations costs and DC system costs which are based on the total installed capacity of the generation system of the energy farm. Although the combined farm decreases the O&M costs of the WTs significantly (being only 67 units and sharing O&M costs with WEC units), the total OPEX (TotOPEX) increases due to the higher costs of the WEC units. This slight increase in the above-mentioned costs is less significant when compared to the decrease of ESS CAPEX in the combined offshore farm. Overall, the combined energy farm not only has merits in reducing ESS capacity requirement while maintaining energy dispatchability to local demand but also presents higher economic feasibility for those locations having good potential for hybrid exploration.

Furthermore, the breakdown costs of the same offshore energy farm in the North sea are shown in the appendix (Fig. A.1). Due to the short transmission distance and shallow water, the SS CAPEX is much lower than that of the Sydney location. However, the cost saving of ESS between wind-only and combined energy farms is lower than that of the Sydney location (17% versus 32%), resulting from different synergies between wind and wave resources, which is consistent with previous analysis.

5. Discussion

Due to the continued retirement of conventional fossil fuel generation and increasing penetration of renewable generation in the modern grid and energy market, ESS play a critical role in providing energy supply firmness to alleviate the uncertainty inherent in renewable resources. In addition, the exploitation of offshore renewables is anticipated on an enormous scale in the near future. This endeavour requires an appropriate assessment of energy storage technologies that can be collocated with ocean energy farms.

Table 3

Technical and economic parameters of main ESS types [52–58].

ESS Types		Volumetric Power Density, kW/m ³	Volumetric Energy Density, kWh/m ³	Gravimetric Power Density, W/kg	Gravimetric Energy Density, Wh/kg	Suitable Storage Duration	Maturity	Efficiency %	UCP, UCE €/kW, €/kWh	O/M €/kw-yr	Life Years Avg.
Mech.	FES	1000–2000	20–80	400–1500	5–100	secs-mins	precommercial	90–95	2765/11059	19	20
	CAES	0.04–10	0.4–20	2.2–24	10–30	hours-months	commercial	42–54	1602/101	21	25
	PHS	0.5–1.5	0.5–2	0.4–1.5	0.5–2	hours-months	mature	70–80	2533/158	3	25
Elec.	Lead–acid	10–400	25–90	75–300	30–50	mins-days	mature	70–90	442/419	48	5
	VRFB	2.5–33	10–33	80–150	30–50	hours-months	precommercial	75–85	432/715	67	20
	Li-ion	56–800	94–500	150–200	75–200	secs-hours	commercial	75–95	373/357	10	12.5
	NaS	140–180	150–300	90–230	150–240	secs-hours	commercial	80–90	432/762	77	15

Table 3 summarizes the major technical and economic parameters of different ESS types, including flywheel energy storage (FES), compressed air energy storage (CAES), pumped hydro storage (PHS), battery electrical storage (BES) options such as lithium-ion (Li-ion), vanadium redox flow batteries (VRFB), lead-acid batteries and Sodium-sulfur (NaS) batteries. Note that the values of parameters are estimated by recent studies, and some parameters may be dependent on specific applications. Based on the key criteria proposed in Section 2.5, the feasibility of these ESS to be integrated with offshore energy farms are examined for the proposed onshore and offshore applications (as shown in Fig. 3 option A and B).

5.1. Evaluation of storage technologies

The volumetric and gravimetric power and energy density are the most important technical parameters when considering the deployment of ESS for an offshore energy farm. These parameters provide fundamental information on the physical size requirement for a given storage capacity, which is critical for offshore deployment. For example, the volume of the ESS is sensitive to the available space on the offshore platform, while the mass of the ESS significantly impacts the dynamics of a floating offshore platform. According to Table 3, FES has the best power density than other ESS with up to 2000 kW/m³ and 1500 W/kg of volumetric and gravimetric power density, while the energy densities are relatively low. Therefore, it is more beneficial for power smoothing [61] in offshore conversion systems instead of energy storage. In addition, considering the large-scale ESS necessary for the defined system, placing large and heavy systems offshore is not feasible. Thus, systems such as CAES, PHS, and VRFB could not be a feasible option to be co-located offshore with generation systems. Note that Lead-acid, Li-ion and NaS batteries generally have better power and energy density in terms of both volumetric and gravimetric aspects.

The storage duration varies based on technology, with some systems providing short-duration storage of seconds to minutes, such as FES and Li-ion and NaS batteries, for power smoothing, regulation and alleviation of ramping events [13], and others offering long-duration storage of hours to days, such as CAES, PHS and VRFB, for balancing the energy demands and dispatchability from renewable energy generation [62]. The choice depends on the application's needs and project goals, such as energy resource availability and local demand characteristics. In terms of enhancing energy supply firmness, the storage duration generally requires a few hours to a day (typical load cycle), eventually resulting in a large energy capacity. In light of this, long storage duration ESS – which are necessary for the above described application – may not be feasible for the offshore deployment.

The ESS efficiency, known as round-trip efficiency, is a general metric that can be used to evaluate energy loss and degradation (related to operation and lifetime). In general, lower efficiency leads to higher energy losses, and larger ESS systems need to be designed to fulfil the same requirements which is a significant drawback for an offshore deployment. From efficiency statistics, as shown in Table 3, most ESS have a similar efficiency range between 70%–95%, except for the CAES (with a lower efficiency between 42%–54%), as ESS efficiency are the technology-specific such as inverters and energy management and

optimization. Note that energy efficiency is not included in the optimal sizing approach, which can be the future extensions of the current study.

In addition, the lifetime of an ESS significantly impacts the total project investment by including high replacement costs. This characteristic becomes more critical for the ESS to be deployed offshore, due to the high cost of replacement activities and limited access window. Offshore wind farms are generally designed to be operated for 20 years, thus, BES's relatively short lifetime (less than the economic life of energy farms) leads to a major challenge for offshore applications. This factor has been integrated into the economic assessment in the next section in the CAPEX and DCPEX values of the lifecycle costs.

The technical maturity of ESS refers to the level of development, reliability, and readiness of these systems for practical use in various applications. It also significantly impacts investment cost and O&M cost for an offshore energy farm, especially when considering the lack of experience in offshore applications. Typically, the high maturity ESS (such as Li-ion battery) is preferred as it is modularized and easier to produce and deploy and might gain financial advantages from up-scaled manufacturing. Also PHS is a mature technology, but it has higher requirements on the geographical location specification. The costs and comparison will be more thoroughly assessed in the following section by two case studies.

It is noted that the actual design, construction and deployment of the ESS is not taken into consideration in this assessment. Regarding the volumetric and gravimetric dimensions of the sized ESS, at a later design stage, it is possible to determine whether a centralized or distributed ESS would be more beneficial. An offshore and distributed ESS would possibly decrease the transmission cable capacity necessary to connect the offshore RES to the onshore grid, smoothen the output of each generation unit that is directly connected to an ESS unit, and reduce the power intermittency in the offshore transmission cables.

5.2. Economic feasibility of storage technologies

Based on the technical evaluation matrix and economic modelling results, the feasibility of different ESS types is comprehensively summarized in Table 4, where 'N/A' denotes non-applicability due to technical infeasibility. The LCOEs of the wind-only energy farm (with ESS) and the combined energy farm (with ESS) are compared in the two locations (Site A and B). It can be seen that the combined energy farm with ESS generally shows a lower LCOE than that of the wind-only farm with ESS in both locations, regardless of the chosen ESS type. In addition, Site A has better synergies for a combined energy farm as more significant cost savings can be made compared to Site B. These are consistent with the discussion in Section 4.3. Furthermore, it is found that due to the modularized design and mature system components, the Li-ion battery can be placed in both on- and off-shore locations. Differently, the CAES mechanical storage only being located at the onshore position shows competitive LCOE values compared to other ESS types in both locations and system configurations. For example, in the Sydney location (site A), the lowest LCOE of 109.8 €/MWh can be found in a combined energy farm by selecting the conventional large-scale CAES onshore, which is closely followed by the Li-Ion battery with the value of 110.4 €/MWh.

Table 4
Techno-economic comparison of ESS in a combined energy farm.

ESS Types		Mechanical Storage			Battery Storage			
Position	Feasibility	FES	CAES	PHS	Li-ion	VRFB	Lead-acid	NaS
Offshore	Technical	×	×	×	✓	×	✓	×
	Economic	N/A	N/A	N/A	✓	N/A	×	N/A
Onshore	Technical	×	✓	✓ or ×	✓	✓	✓	✓
	Economic	N/A	✓	✓	✓	✓	×	×
Locations	Configurations	LCOE (€/MWh)						
Site A	WT+ESS:	N/A	126.6	140.8	132.0	166.7	180.7	176.8
	WW+ESS:	N/A	109.8	119.3	110.4	131.5	139.1	137.5
Site B	WT+ESS:	N/A	121.5	137.3	125.1	162.0	176.1	172.6
	WW+ESS:	N/A	115.6	128.1	118.0	146.7	157.6	155.1

Note that, unlike the Li-ion battery, the CAES does not offer technical flexibility (like modularized and mature system components) to deploy on an offshore platform. Although some small-scale CAES are being developed and could be available for offshore applications, such technologies are more expensive than traditional large-scale CAES, and their costs are currently hard to predict due to low maturity [54]. Furthermore, the impact of energy efficiency on the ESS capacity and on the total energy generated is not quantitatively considered in the financial assessment as it depends on the specific design (such as scales and implementations) of the chosen ESS and on its application.

Note that CAES is a more cost-effective solution for large-scale energy storage applications compared to Li-ion batteries due to its low energy capital cost and long lifetime. Lead-acid and NaS batteries are the most expensive technologies to date for both energy farm configurations and locations because of their short lifetime and high O&M costs. It is worth noting that PHS has the economic and technical potential to be integrated with large-scale energy farms for long-duration energy storage when a suitable nearshore location is available.

Finally, no offshore ESS technology currently can offer an acceptable LCOE value (e.g. lower than their onshore counterparts) to justify the high cost of offshore installation and operation and display economic opportunities. In this paper, costing values have been obtained from publicly available data, which did not include offshore-specific information. However, in future studies, an in-depth analysis of the cost advantages of the combined energy farm with an offshore storage system will be studied to analyse the approach used in this paper.

6. Conclusion

In this work the ESS requirements in offshore wind and combined wind and wave energy farms are studied by developing a novel statistical method, which considers the reduction of the energy variability of RES while meeting an energy dispatchability to the local energy demand. The cost models of different offshore wind farm configurations are developed to investigate the economic synergies among the combined energy farms. In addition, the feasibility of different ESS types is comprehensively discussed and assessed based on the proposed multiple technical and economic criteria. Furthermore, the sensitivity of location-specific characteristics on the benefits of the combined energy farms and ESS capacity requirements is analysed.

The results indicate that, compared to the stand-alone wind energy farm, the combined wind and wave energy farm can significantly reduce the storage capacity (with power capacity up to 20% and energy capacity up to 35%) to meet the energy dispatch commitment to the local demand, hence decreasing the LCOE. This provides a straightforward and practical justification for understanding the energy variability reduction of combined offshore farms instead of previous studies purely focusing on analysing the variability index. In addition, this research offers the foundational tool to provide the quantitation analysis of the techno-economic benefits of combined energy farm configuration,

an aspect that remains unaddressed by the limitations of the energy variability index.

It is noted that the economic benefits of a combined energy system vary with the location characteristics, including sea sites with low correlation and long lag time between wind and wave, hence they have a lower LCOE and vice versa. Regarding ESS technologies, Lithium-ion batteries have a better overall performance in terms of applicability and techno-economic feasibility. However, other energy storage technologies, such as NaS batteries, are also found promising.

Note that interactions between WT and WEC and the direction of the wind is assumed negligible when assessing the energy generated of the RES. In addition, due to the immature WEC technology (insufficient experience) and large uncertainties in offshore storage applications, the LCC model of the WEC and ESS sub-systems does not provide accurate values for the financial metric. Moreover, efficiency and detailed deployment of offshore ESS in relation to the installation and operation are not considered in this study as they are project-specific. Future work will address these limitations and investigate the revenue model of the combined energy farm with ESS while considering the energy efficiency and usage of the storage system.

CRedit authorship contribution statement

Qiang Gao: Conception, Methodology, Analysis, Drafting. **Alva Bechlenberg:** Methodology, Analysis, Validation, Drafting. **Bayu Jayawardhana:** Supervision, Knowledge, Review & editing. **Nesimi Ertugrul:** Supervision, Knowledge, Review & editing. **Antonios I. Vakis:** Supervision, Knowledge, Analysis, Review & editing. **Boyin Ding:** Supervision, Knowledge, Analysis, Review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix. Figures

See Fig. A.1.

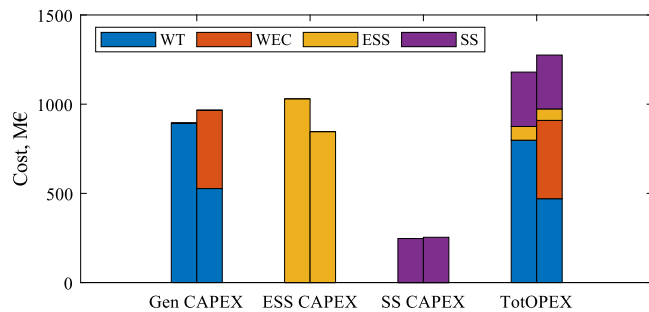


Fig. A.1. Breakdown of undiscounted costs of 500 MW offshore energy farms in the North Sea location, under wind alone (left groups) and combined system (right groups) configurations.

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